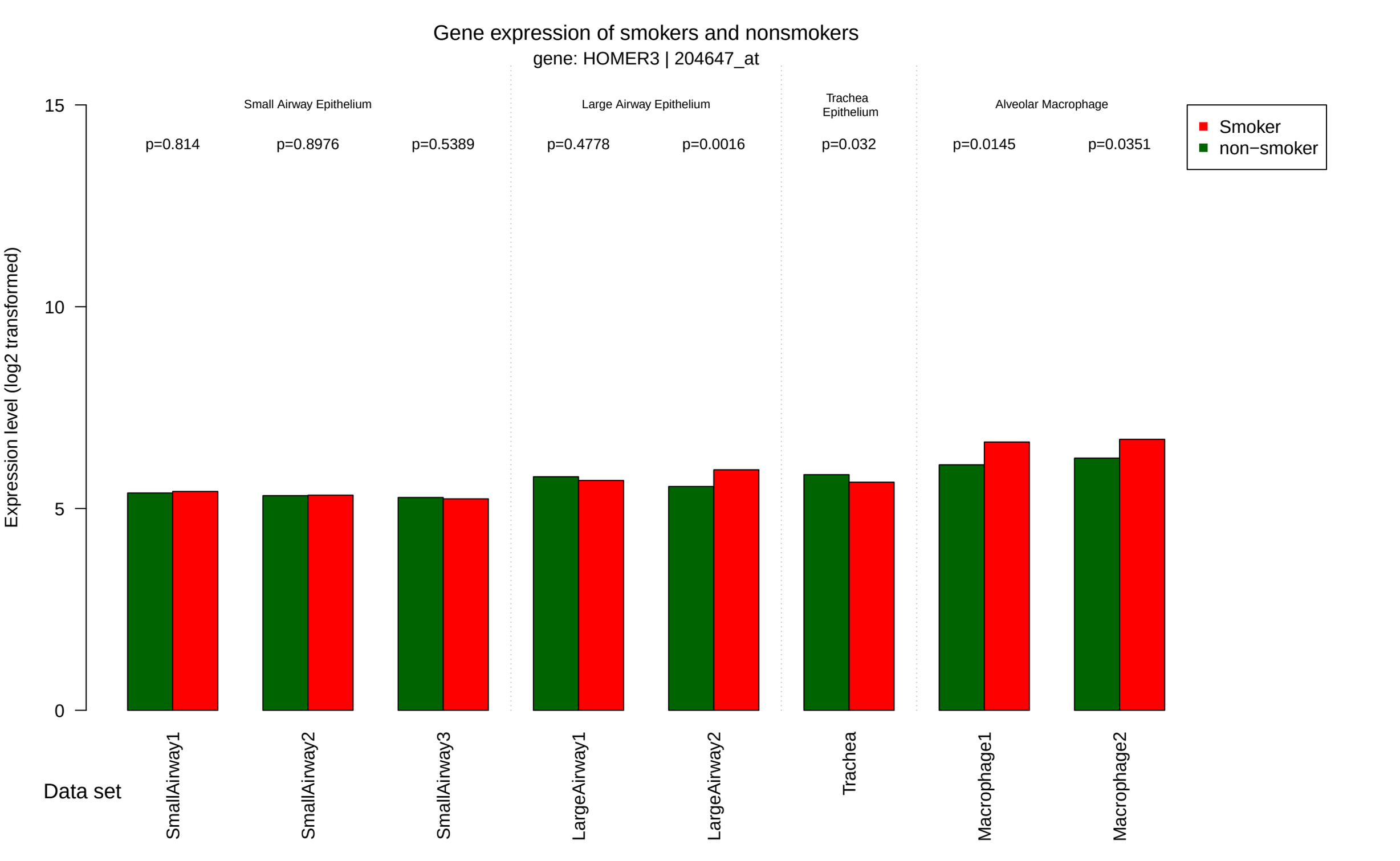
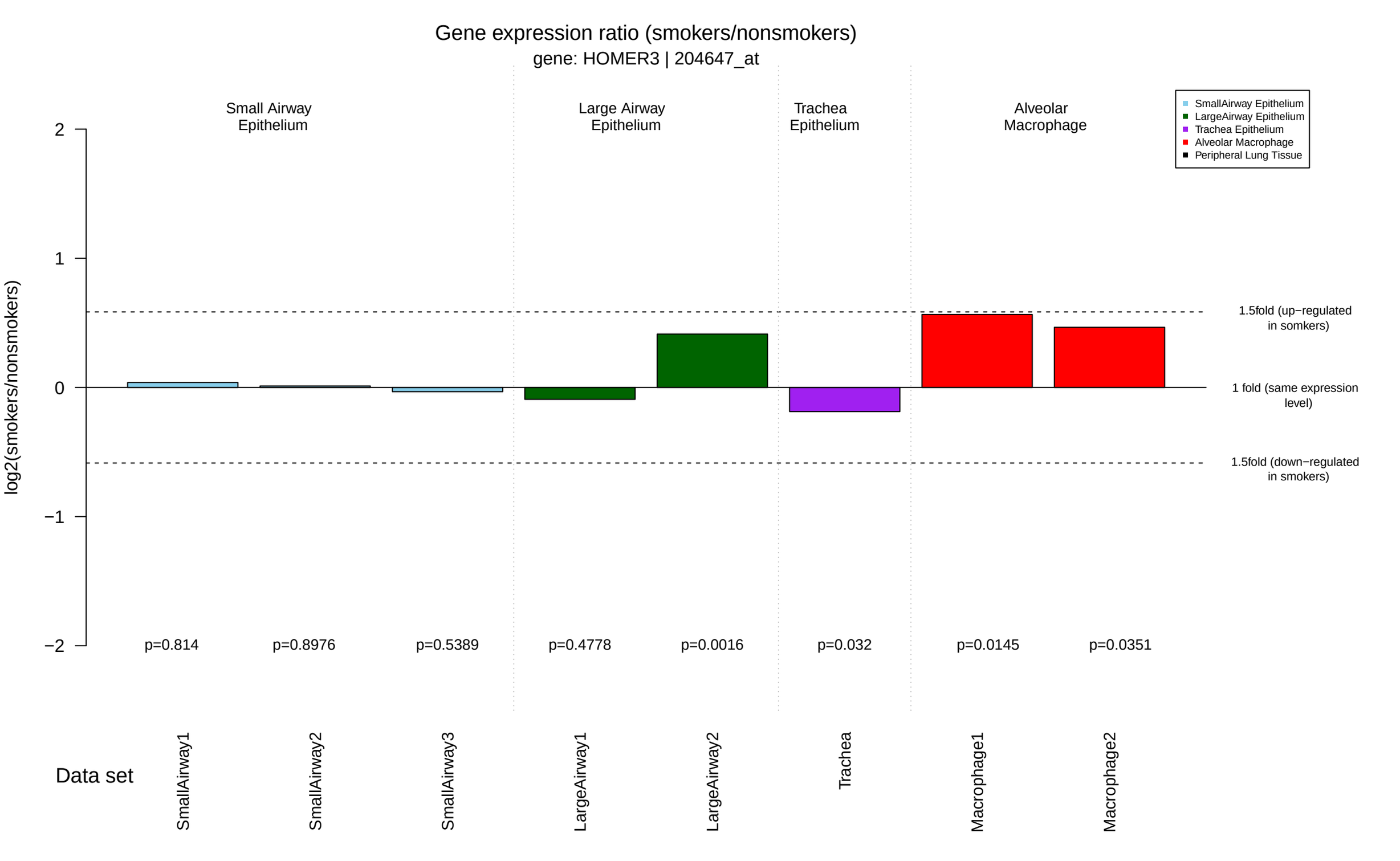
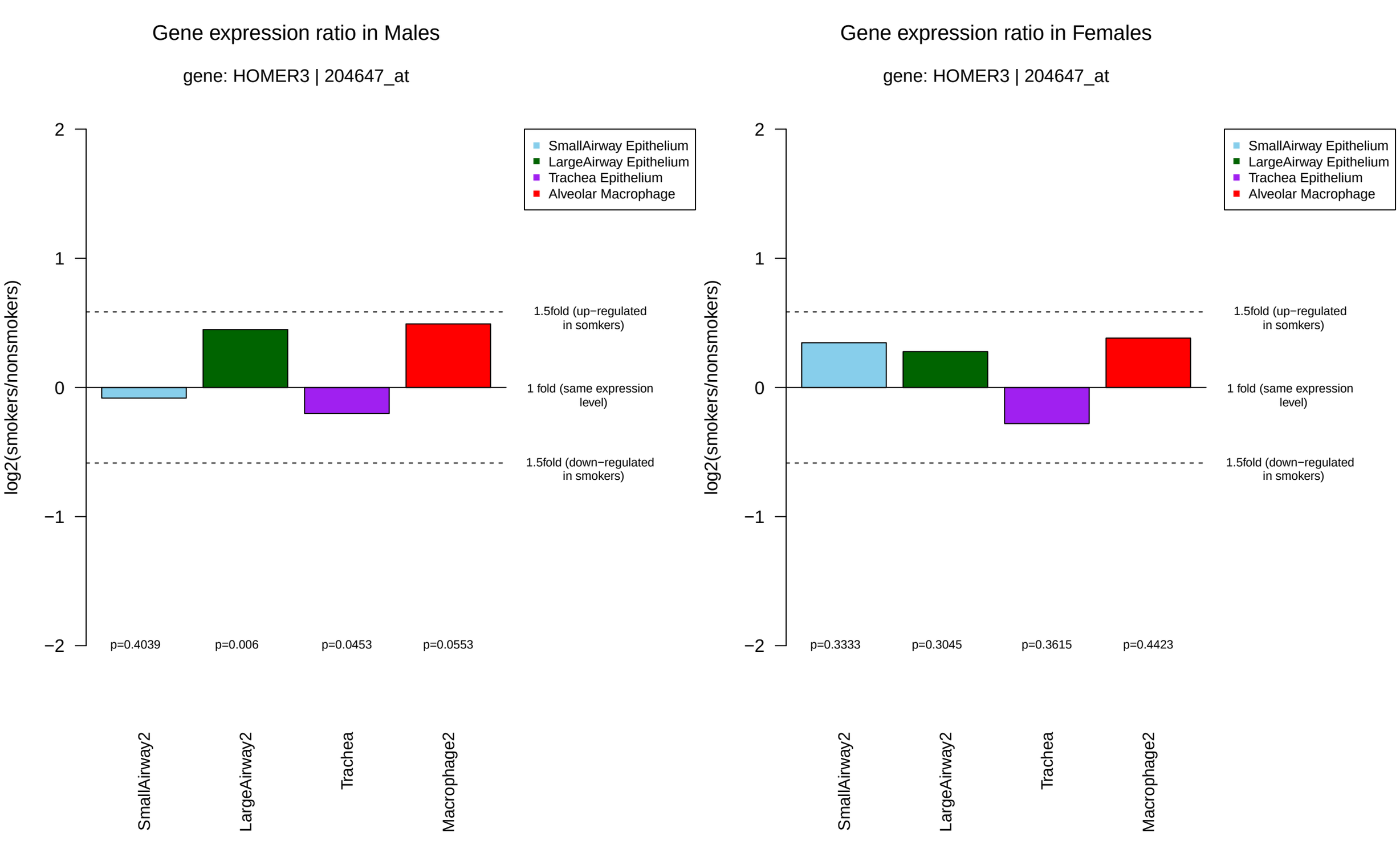
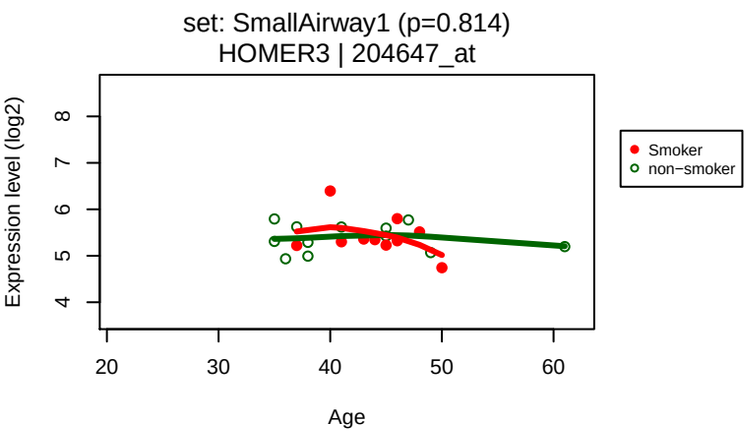
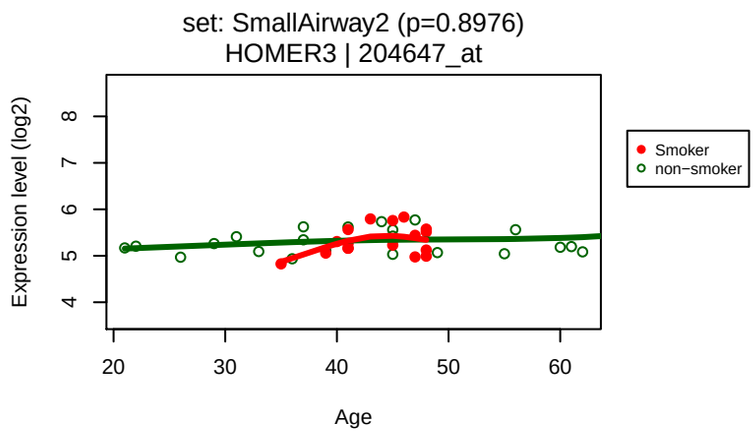
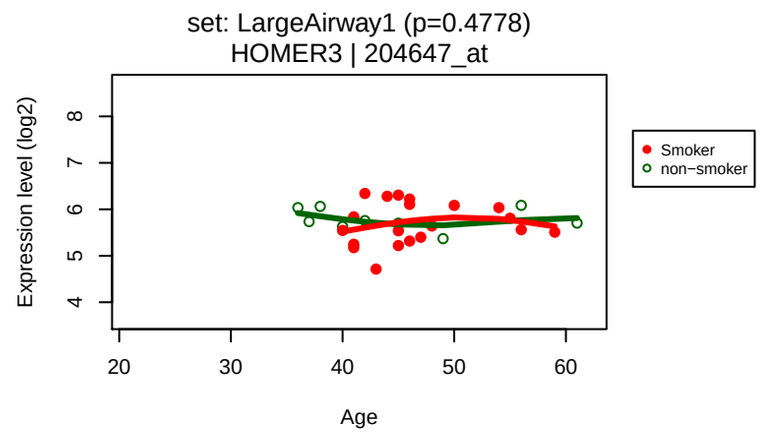

Gene expression of smokers and nonsmokers
gene: HOMER3 | 204647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.814
p=0.8976
p=0.5389
p=0.4778
p=0.0016
p=0.032
p=0.0145
p=0.0351
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

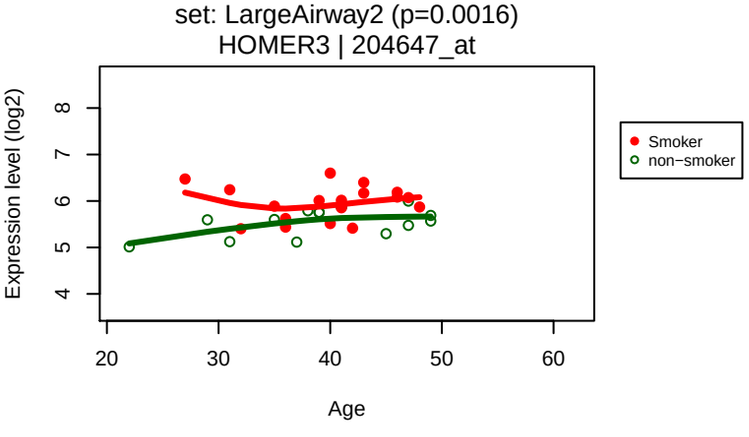

Gene expression ratio (smokers/nonsmokers)
gene: HOMER3 | 204647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.814
p=0.8976
p=0.5389
p=0.4778
p=0.0016
p=0.032
p=0.0145
p=0.0351
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

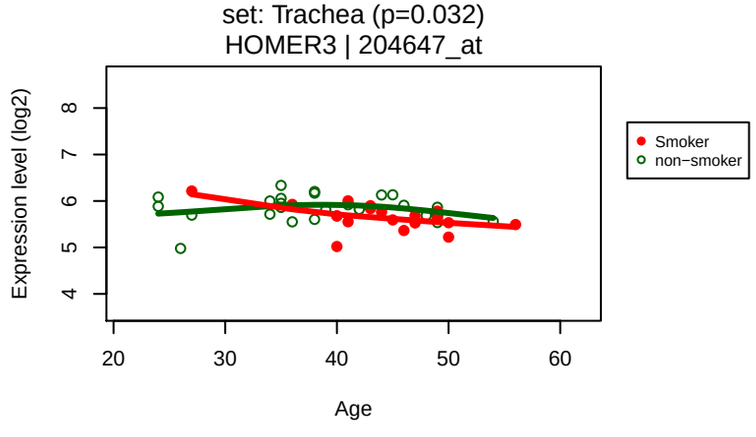

Gene expression ratio in Males
gene: HOMER3 | 204647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4039
p=0.006
p=0.0453
p=0.0553
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOMER3 | 204647_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3333
p=0.3045
p=0.3615
p=0.4423
SmallAirway2
LargeAirway2
Trachea
Macrophage2

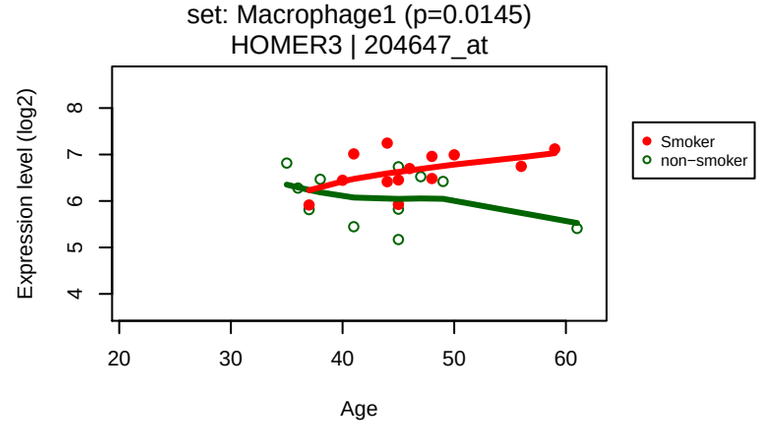

set: SmallAirway1 (p=0.814)
HOMER3 | 204647_at
Expression level (log2)
Age
Smoker
non−smoker

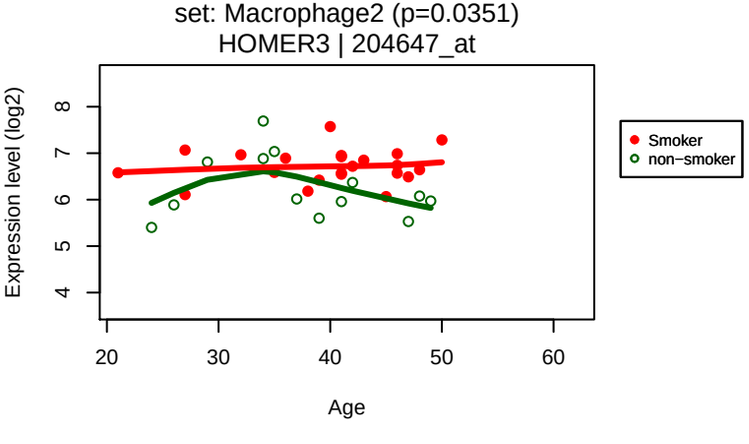

set: SmallAirway2 (p=0.8976)
HOMER3 | 204647_at
Expression level (log2)
Age
Smoker
non−smoker

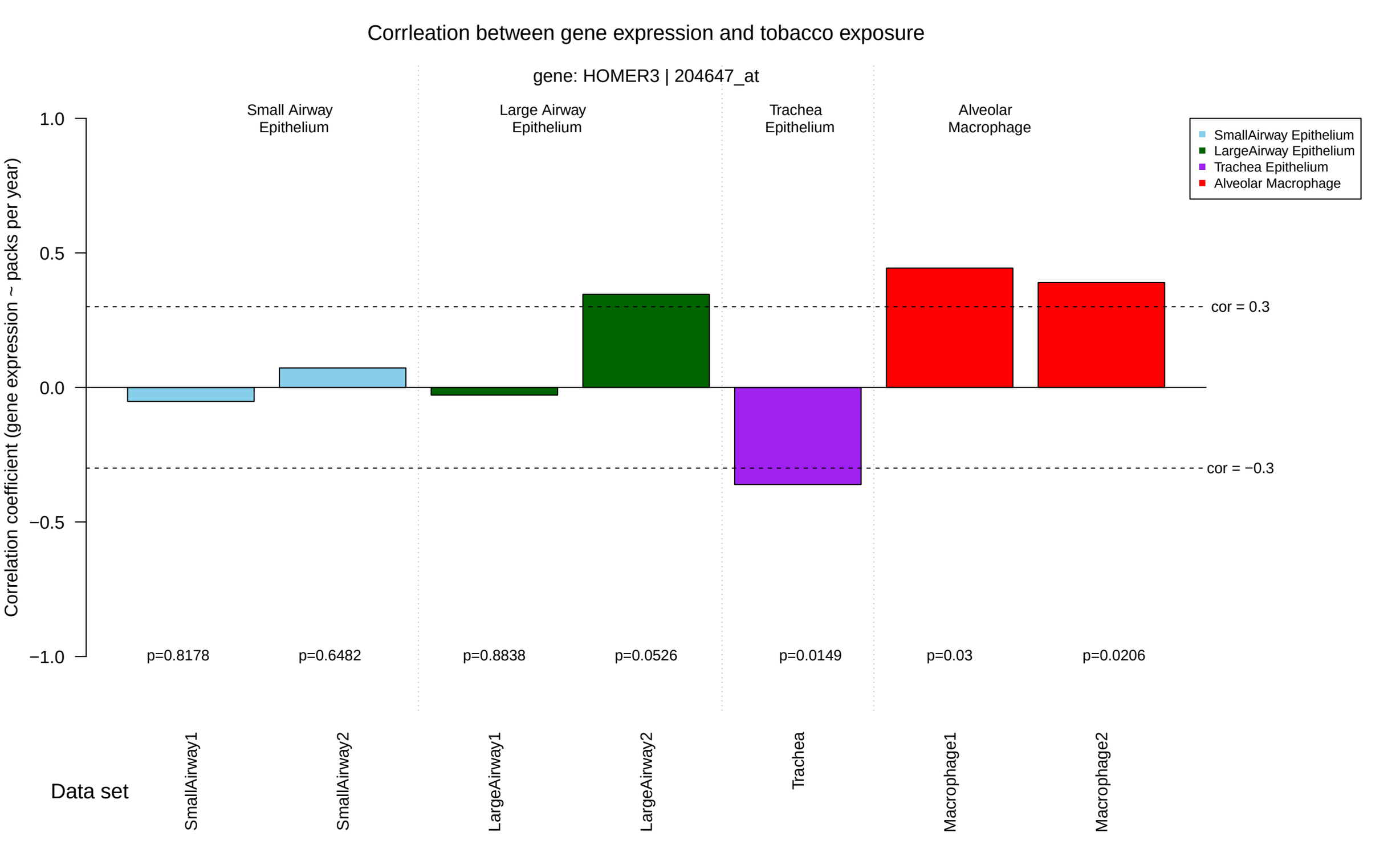

set: LargeAirway1 (p=0.4778)
HOMER3 | 204647_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0016)
HOMER3 | 204647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.032)
HOMER3 | 204647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0145)
HOMER3 | 204647_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0351)
HOMER3 | 204647_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOMER3 | 204647_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8178
p=0.6482
p=0.8838
p=0.0526
p=0.0149
p=0.03
p=0.0206
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2